

8 Required Baseline Propositions

This section states exactly six substantive propositions. The novelty result is a corollary of the MAH-relevant-project-set identity, not a seventh substantive proposition. All policy comparisons are finite comparisons between $M = 0$ and $M = 1$; the binary regime is never differentiated.

8.1 Proposition 1: Organizational sorting

Assumptions and fixed objects. Fix a project (q, m) , a finite institutional wedge, regime $M = 1$, and a conjectured support price $p_m < \infty$. Hold all primitives except internal manufacturing capability k_i fixed. On the interior of the internally feasible domain, use the technology signs and endpoint-crossing conditions stated above.

Claim. The internal-minus-entrusted value gap is strictly increasing in k_i :

$$\frac{\partial \Delta_{IE}(k_i; q, m, M, p_m)}{\partial k_i} = s(q)R_c(q, c_I)c_{I,k}(m, k_i) - F_{I,k}(m, k_i) > 0. \quad (1)$$

If the gap is negative at the lower feasible endpoint and positive at the upper endpoint, there is a unique $k^*(q, m; p_m, M)$. Conditional on both I and E dominating T and A , developers below the cutoff use E and developers above it use I .

Proof. The commercial-value kernel gives $R_c < 0$, while the internal technology gives $c_{I,k} < 0$ and $F_{I,k} < 0$. The operating-cost term on the right side of (1) is weakly positive (and is zero when $s(q) = 0$), while the setup-cost term is strictly positive. Their sum is therefore strictly positive. Continuity and endpoint crossing give existence by the intermediate value theorem; strict monotonicity gives uniqueness. The sign of the gap on either side of its unique zero gives the conditional sorting statement. The comparison with T/A is required because an I/E ranking alone does not establish the globally selected route.

Sufficient conditions and zero-effect cases. The endpoint-crossing conditions are sufficient, not primitive conclusions. If they fail, a finite interior cutoff need not exist. If $k_i < \underline{k}(m)$, internal production is infeasible. As $k_i \rightarrow \infty$, the maintained technology signs weakly favor internal production relative to entrusted production. At $M = 0$, route E is unavailable and there is no finite pre-MAH I/E cutoff.

Economic interpretation. Internal capability reduces both internal operating cost and setup burden. Entrusted production is most useful to developers for which those internal advantages are weak, but it is selected only when it also beats transfer or delay. The result is organizational sorting, not a permanent firm-type taxonomy.

8.2 Proposition 2: The MAH-relevant project set

Assumptions and fixed objects. Fix developer i , project draw (q, m) , and support price p_m . The old route set is I, T, A ; post-MAH adds E without removing an old option. Route values retain the definitions given above.

Definitions and claim. Define

$$W_i^0(q, m) = \max\{W_i^I(q, m), W_i^T(q, m), 0\}, \quad W_i^1(q, m; p_m) = \max\{W_i^0(q, m), W_i^E(q, m; 1, p_m)\}. \quad (2)$$

Then

$$W_i^1(q, m; p_m) - W_i^0(q, m) = \left[W_i^E(q, m; 1, p_m) - W_i^0(q, m) \right]_+. \quad (3)$$

The MAH-relevant project set is

$$\mathcal{C}_i(p_m) = \left\{ (q, m) : W_i^E(q, m; 1, p_m) > W_i^0(q, m) \right\}. \quad (4)$$

Only project draws in this set receive a strict direct value gain at the fixed support price.

Proof. For any real numbers a, b , $\max\{a, b\} - a = [b - a]_+$. Apply the identity with $a = W_i^0$ and $b = W_i^E$. The gain is strictly positive exactly where the argument of the positive-part operator is positive, which is precisely $\mathcal{C}_i(p_m)$.

Sufficient conditions and zero-effect cases. A project receives a strict gain if and only if entrusted value strictly exceeds internal value, transfer value, and abandonment value at the support price. If $W_i^E \leq W_i^0$, the project gain is zero. If this inequality holds for all project draws, $\mathcal{C}_i(p_m)$ is empty up to null sets. As $p_m \rightarrow \infty$, $W_i^E \rightarrow -\infty$, so the old value is preserved in the limit for each fixed project. Each fixed project eventually exits the relevant set; without bounded support this need not occur at one common finite price. The integrable value envelope gives convergence of expected gain to zero.

Economic interpretation. MAH matters only where retained entrusted manufacturing beats every option already available. The reform does not raise the value of projects that were already better served internally, transferred, or delayed.

8.3 Proposition 3: Project advancement and firm heterogeneity

Assumptions and fixed objects. Fix a support price p_m , policy-invariant primitives, and developer characteristics except for the characteristic varied in each comparison. Route values do not depend on research capability a_i . For the manufacturing-capability result, additionally impose the proposition-specific sufficient condition $\nu \geq 1$. For comparisons including the exact feasibility endpoint, require right-continuity of feasible internal costs there; no derivative across the extended-value boundary is taken.

Proof: expected value and advancement response. Let $\Omega_i^h(p_m) = \int W_i^h(q, m; p_m) dF(q, m)$ for $h \in \{0, 1\}$, with the irrelevant p_m argument suppressed for $h = 0$. Proposition 2 implies

$$\Delta\Omega_i(p_m) \equiv \Omega_i^1(p_m) - \Omega_i^0 = \int \left[W_i^E(q, m; 1, p_m) - W_i^0(q, m) \right]_+ dF(q, m) \geq 0. \quad (5)$$

The common advancement response is

$$\Delta x_i(p_m) = \left(\frac{\beta a_i}{\kappa} \right)^{1/\nu} \left[\left(\Omega_i^0 + \Delta\Omega_i(p_m) \right)^{1/\nu} - \left(\Omega_i^0 \right)^{1/\nu} \right]. \quad (6)$$

Because $t^{1/\nu}$ is strictly increasing for every $\nu > 0$,

$$\Delta x_i(p_m) > 0 \iff \Delta\Omega_i(p_m) > 0.$$

Research-capability scaling. Holding $k_i, p_m, \Omega_i^0, \Delta\Omega_i$ and all primitives fixed, a positive response satisfies

$$\frac{\partial \Delta x_i(p_m)}{\partial a_i} = \frac{\Delta x_i(p_m)}{\nu a_i} > 0. \quad (7)$$

Thus the level response is stronger at higher a_i whenever the MAH-relevant set has positive expected surplus.

Capability result. The internal value is nondecreasing in k_i , whereas the other old route values and entrusted value are independent of k_i . Therefore $\Omega_i^0(k_i)$ is nondecreasing and $\Delta\Omega_i(k_i; p_m)$ is nonincreasing. When $\nu \geq 1$, $h(t) = t^{1/\nu}$ is increasing and concave, so $h(u+d) - h(u)$ is nondecreasing in d and nonincreasing in u . Applying this result to $u = \Omega_i^0(k_i)$ and $d = \Delta\Omega_i(k_i; p_m)$ yields

$$k'_i \geq k_i \implies \Delta x_i(k'_i; p_m) \leq \Delta x_i(k_i; p_m), \quad \nu \geq 1. \quad (8)$$

The inequality is strict when the entrusted surplus falls on a positive-measure set. For $\nu > 1$, a strict rise in baseline value also makes the concavity channel strict when the reform surplus remains positive; zero surplus cannot generate a strict response. For $0 < \nu < 1$, no sign is asserted without an additional bound.

Zero-effect and boundary cases. If the relevant set is null or entrusted advantage is zero, then $\Delta\Omega_i = \Delta x_i = 0$. At $\nu = 1$,

$$x_i^* = \frac{\beta a_i}{\kappa} \Omega_i, \quad \Delta x_i = \frac{\beta a_i}{\kappa} \Delta\Omega_i,$$

so manufacturing heterogeneity operates entirely through the expected value gain. A high a_i does not create a response when the value gain is zero.

Economic interpretation. Research capability scales how strongly a given increase in the value of advancing projects changes the level of advancement. Weak internal manufacturing capability makes retained outsourcing more likely to add value. The resulting response is strongest for high- a_i , low- or intermediate- k_i developers under the stated curvature condition. This does not imply more patent applications, basic research, scientific discovery, or breakthrough novelty.

8.4 Corollary: Novelty composition is ambiguous

Assumptions and fixed objects. Fix developer i and support price p_m . The class $g \in \{O, \text{Inc}\}$ is an exogenous classifier, not a control or state. The same x_i applies to both classes.

Claim and proof. Define $\Omega_{ig}^0 = E_{F_g}[W_i^0]$ and $\Omega_{ig}^1 = E_{F_g}[\max\{W_i^0, W_i^E\}]$. Then

$$\Delta\Omega_{ig} = E_{F_g} \left[\left(W_i^E - W_i^0 \right)_+ \right], \quad \Delta\Omega_i = \sum_{g \in \{O, \text{Inc}\}} \rho_g \Delta\Omega_{ig}. \quad (9)$$

The positive-part identity proves the first equality; the mixture identity for F proves the second. The baseline imposes no restriction ordering the two conditional distributions of entrusted surplus. A support on which only class O has positive entrusted surplus gives $\Delta\Omega_{iO} > \Delta\Omega_{i\text{Inc}}$; interchanging the class supports reverses the ranking. Hence neither ordering follows from the baseline.

Zero-effect and boundary cases. If $\rho_O = 0$, class O contributes zero to the aggregate gain; if $\rho_{\text{Inc}} = 0$, class Inc contributes zero. A conditional class gain may be zero even when the other is positive.

Economic interpretation. Different novelty classes may induce different distributions of manufacturing requirements, commercial values, and hence success probabilities, costs and outside options through their F_g distributions. Only explicit primitive differences can order their responses. No empirical original-versus-incremental ranking is hard-coded into the theory.

8.5 Proposition 4: CMO equilibrium existence and uniqueness

Assumptions and fixed objects. Fix M and all exogenous distributions and primitives. Use the maintained primitive supply, background-demand, no-tie and integrability conditions of the capacity-market section. That section derives continuous nonincreasing finite demand and continuous strictly increasing supply, with positive excess demand at zero and negative excess demand at high price. These aggregate properties are derived inputs to this proof, not assumed economic conclusions.

Claim. There exists a unique qualified-capacity clearing price $p_m^*(M) > 0$.

Proof. Define excess demand

$$Z_M(p_m) = D_m(p_m; M) - S_m(p_m). \quad (10)$$

It is continuous. At zero, $Z_M(0) \geq D_m^B(0) > 0$. At sufficiently high prices, total demand tends to zero and supply is unbounded, so $Z_M(p_m) < 0$. The intermediate value theorem gives a positive root. Because demand is weakly decreasing and supply strictly increasing, Z_M is strictly decreasing; the root is unique.

Sufficient conditions and zero-effect cases. Strictly increasing supply is sufficient for uniqueness even when demand has flat regions. At $M = 0$, study-related entrusted demand is zero, but background demand supports a positive pre-MAH price. If background demand at zero were absent, the positive-price conclusion would not follow from these conditions; that is outside the maintained baseline.

Economic interpretation. The capacity price coordinates deterministic entrusted-route demand, project advancement, and supplier capacity through one scalar market. Its existence does not require logit smoothing, supplier entry, or another market.

8.6 Proposition 5: CMO scarcity attenuation

Assumptions and fixed objects. Let $p_m^0 = p_m^*(0)$ and $p_m^1 = p_m^*(1)$. Hold qualified-supply technology, supplier distribution and background demand fixed across regimes. The reform adds nonnegative entrusted-route study demand; old routes remain available.

Proof: equilibrium price comparison. At the pre-MAH price,

$$Z_1(p_m^0) = D_m^{\text{MAH}}(p_m^0; 1) \geq 0,$$

because background demand equals supply at p_m^0 . Since Z_1 is strictly decreasing and vanishes at p_m^1 ,

$$p_m^1 \geq p_m^0, \quad \text{with strict inequality if } D_m^{\text{MAH}}(p_m^0; 1) > 0. \quad (11)$$

Value attenuation. Define the fixed-price and equilibrium-price gains by

$$\Delta\Omega_i^{\text{dir}} = \int [W_i^E(q, m; 1, p_m^0) - W_i^0(q, m)]_+ dF,$$

$$\Delta\Omega_i^{\text{eq}} = \int [W_i^E(q, m; 1, p_m^1) - W_i^0(q, m)]_+ dF.$$

Since $W_{i,p_m}^E = -b(m) < 0$ and $p_m^1 \geq p_m^0$, the integrand in the equilibrium expression is pointwise no larger. Both integrands are nonnegative, so

$$0 \leq \Delta\Omega_i^{\text{eq}} \leq \Delta\Omega_i^{\text{dir}}. \quad (12)$$

Advancement attenuation. For a fixed pre-MAH value Ω_i^0 , the mapping

$$d \mapsto \left(\frac{\beta a_i}{\kappa} \right)^{1/\nu} [(\Omega_i^0 + d)^{1/\nu} - (\Omega_i^0)^{1/\nu}]$$

is nonnegative and strictly increasing for $d \geq 0$. Therefore

$$0 \leq \Delta x_i^{\text{eq}} \leq \Delta x_i^{\text{dir}}. \quad (13)$$

Zero-effect and boundary cases. If study demand is zero at p_m^0 , uniqueness gives $p_m^1 = p_m^0$. If supply is perfectly elastic at a common $\bar{p}_m$, then $p_m^1 = p_m^0 = \bar{p}_m$ and attenuation disappears. If a high equilibrium price eliminates entrusted advantage for developer i , its equilibrium gain is zero. Even then the old route-set value is preserved: scarcity can attenuate the positive reform gain but cannot make access to the enlarged choice set privately worse.

Economic interpretation. New demand for qualified external capacity bids up its scarcity price. This reduces the value of the entrusted option relative to a fixed-price policy calculation and tempers project advancement. The price response is endogenous; MAH does not directly reduce scarcity or shift supply.

8.7 Proposition 6: Planning-stage and observed outcomes

Assumptions and fixed objects. Fix a regime and its equilibrium objects. The common x_i^* is selected before project characteristics and route assignment. The downstream probability $s(q)$ is exogenous and policy invariant. For typed baseline outcomes write $s_g(q) = s(q)$: the same kernel enters valuation and realized outcomes. The optional separate class-specific kernel is not activated. All integrals use the deterministic route choices evaluated in the same regime.

Planning-stage projects. The equilibrium intensity at which viable projects reach the stage relevant for commercialization is

$$\Lambda_i^{\text{plan}} = a_i x_i^*. \quad (14)$$

This is the planning-stage technology $\lambda_i^{\text{plan}} = a_i x_i$ evaluated at the advancement optimizer.

Observed retained outcomes. Within the paper's original-drug domain, expected realized outcomes for which the developer remains the holder are

$$Y_i^{\text{ret}} = a_i x_i^* \int s(q) [\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}] dF(q, m). \quad (15)$$

Expected realized outcomes with retained holder–producer separation are

$$Y_i^E = a_i x_i^* \int s(q) \mathbf{1}\{r_i^* = E\} dF(q, m). \quad (16)$$

For $g \in \{O, \text{Inc}\}$, the class contribution to retained outcomes is

$$Y_{ig}^{\text{ret}} = a_i x_i^* \rho_g \int s_g(q) [\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}] dF_g(q, m). \quad (17)$$

There is one common x_i^* , not an x_{ig} control. By the mixture identity and the common realization kernel, $\sum_g Y_{ig}^{\text{ret}} = Y_i^{\text{ret}}$.

Proof and exact decomposition. Let

$$Q_i^{\text{ret},h} = \int s(q) [\mathbf{1}\{r_i^{*,h} = I\} + \mathbf{1}\{r_i^{*,h} = E\}] dF$$

and let $x_i^h = x_i^*(h, p_m^h)$. Adding and subtracting $a_i x_i^1 Q_i^{\text{ret},0}$ gives

$$\Delta Y_i^{\text{ret}} = a_i (x_i^1 - x_i^0) Q_i^{\text{ret},0} + a_i x_i^1 (Q_i^{\text{ret},1} - Q_i^{\text{ret},0}). \quad (18)$$

The first term is the project-advancement/planning-arrival component. The second is the retained-route composition component. Reassignment from I to E can raise Y_i^E without changing the retained total; movement from T/A to E can raise both.

Zero-effect and ambiguity cases. If advancement and all route indicators are unchanged, observed outcomes are unchanged. Since old route values remain available and price invariant, a previously retained project can stay on I or switch to E , but cannot switch to a strictly better T/A option after reform. Thus retained realization mass is weakly increasing (away from null ties). Together with Proposition 5 this yields weakly nonnegative retained-outcome changes, while strict increases and cross-class rankings need further conditions. A class-specific outcome can have zero change while the other changes, for example when the unchanged class has zero retained realization mass in both regimes. Zero conditional value gain alone is not sufficient: if common advancement increases and a class has positive pre-reform retained realization mass, that class also receives the advancement component. The baseline allows $\Delta Y_{iO}^{\text{ret}} = 0 < \Delta Y_{i\text{Inc}}^{\text{ret}}$ and the reverse; it imposes no ranking. If $\rho_g = 0$, that class contributes zero to the aggregate.

Economic interpretation. The model separates investment in advancing viable projects, their arrival at the commercialization-relevant planning stage, organizational assignment, and realized products. Holder–producer separation is observed only after route assignment. None of these expressions makes patent generation endogenous or lets MAH directly change downstream realization probabilities.

8.8 Scope conclusion

The six propositions and the novelty corollary exhaust the baseline results established in this section. They add no market, no entry margin, no welfare object, no logit route share, no class-specific advancement control, and no direct policy effect on research capability, project draws, manufacturing technology, capacity supply, or downstream realization.